

Can the frozen encoder of a diffusion model detect deepfakes?

An encoder-only approach, benchmarked against a ResNet baseline

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Outline

Introduction

Problem statement

Method

Experimental setup

Results

Discussion and next steps

Introduction

Problem setting

Deepfakes are facial images synthesised or manipulated by deep generative models (face-swap, reenactment).

The scientific difficulty is **not** in-distribution accuracy but **generalisation**: a detector trained on one manipulation method typically degrades sharply on a different, unseen one.

- Manipulation methods evolve faster than labelled datasets.
- A detector must capture *generation traces*, not dataset-specific cues.
- Cross-dataset evaluation is the appropriate stress test.

Diffusion models as feature extractors

Latent diffusion models (LDMs) embed an image into a latent space via an encoder $\mathcal{E}$. Recent detectors exploit this:

- Wang & Kalogeiton (ECCV 2024): classifier-free guidance residuals across denoising steps.
- Ricker et al. (AEROBLADE, CVPR 2024): autoencoder reconstruction error.

Both rely on the **full diffusion process** — multiple network evaluations per image. We ask whether a far cheaper signal already suffices.

Problem statement

Hypothesis

The latent representation produced by a **frozen** LDM encoder, taken **alone** — no diffusion, no fine-tuning — is sufficient to discriminate real from manipulated faces.

We assess the hypothesis through a **controlled comparison** against a standard frozen feature extractor (ResNet-50):

- identical classifier head, data and protocol;
- only the feature extractor changes;
- **the backbone is frozen in all cases** (project constraint).

Method

Pipeline



Only the MLP is trained. The extractor is frozen, so latents are **pre-computed once** and cached — training reduces to fitting a small MLP on fixed feature vectors.

Two frozen feature extractors

LDM encoder

Stable Diffusion 1.4 VAE encoder
(Wang & Kalogeiton source).

latent $4 \times 64 \times 64$

$\Rightarrow$ 16 384 features

ResNet-50 (baseline)

ImageNet-pretrained.

Global-average-pooled features
(layer before the head).

$\Rightarrow$ 2 048 features

Identical downstream 3-layer MLP with ReLU activations and dropout 0.3. The only variable is the extractor.

Experimental setup

Datasets and protocol

- **In-domain (train/val/test):** FaceForensics++, c23 compression, pre-cropped faces. **Manipulation used:** DeepFakes (one of the four FF++ manipulations available, the other three are reserved for future work).
- **Cross-dataset (test only):** Celeb-DF-v2 — never seen during training; measures generalisation.
- **Splits:** official Rössler 2019 splits — **identity-disjoint**, preventing identity leakage that would inflate the score.
- **Sampling:** 5 frames per video. **Metric:** AUC, with a 95% video-level bootstrap confidence interval.

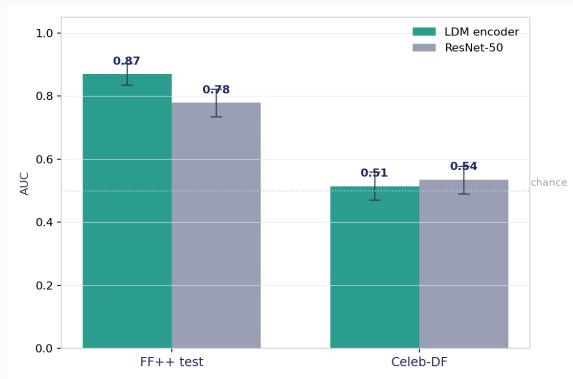
7190 train / 1397 val / 1400 FF++ test / 2590 Celeb-DF test.

Results

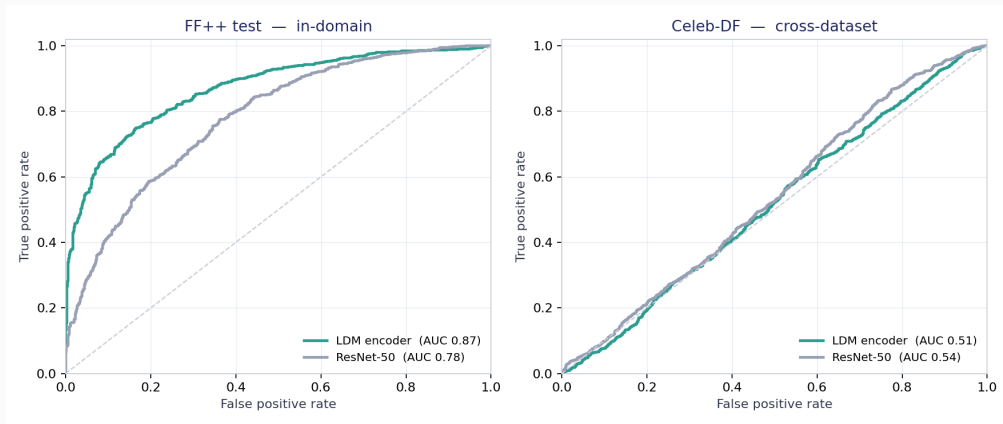
Main comparison (both extractors frozen)

Extractor	FF++	Celeb-DF
LDM encoder	0.87	0.52
ResNet-50	0.78	0.54

AUC with 95% video-level bootstrap CI.
In-domain, the diffusion encoder exceeds the ResNet baseline; cross-dataset, both collapse toward chance (0.5).



ROC analysis



Result

At an equal frozen setup and an identical classifier, the diffusion encoder extracts **more discriminative** features than a standard ResNet: AUC 0.87 [0.835, 0.903] versus 0.78 [0.734, 0.822] on FF++.

- The 95% confidence intervals are **disjoint** (0.835 $\nsubseteq$ 0.822): the in-domain superiority is statistically significant.
- Supports the hypothesis: the encoder-only representation is *not* too weak.

Generalisation gap

Cross-dataset, both extractors fall close to chance: LDM 0.52 [0.471, 0.559], ResNet 0.54 [0.490, 0.577]. Both confidence intervals contain 0.5 and overlap — statistically indistinguishable from chance, and from each other.

Diagnostic

A linear probe trained *directly* on Celeb-DF latents reaches an AUC of about 0.74: the discriminative signal **exists**. The FF++-trained classifier simply fails to transfer.

⇒ This is a **domain gap**, common to both extractors — a limitation of the frozen, single-manipulation setup, not a flaw specific to the diffusion encoder.

Discussion and next steps

Next steps (frozen-backbone compatible)

1. **Improve the MLP head** — explore deeper or wider variants, dimensionality reduction of the latent (pooling), stronger regularisation, hyper-parameter tuning. The bottleneck appears to be the classifier capacity, not the extractor.
2. **Train on the other FF++ manipulations** — currently only DeepFakes is used; adding Face2Face, FaceSwap and NeuralTextures should expose the head to more diverse generation traces and improve cross-method generalisation.

Both steps preserve the frozen-backbone constraint: only the head and the training distribution change.

- The frozen LDM encoder is a **promising** feature extractor: it outperforms a frozen ResNet in-domain (0.87 vs 0.78, confidence intervals disjoint).
- Cross-dataset generalisation is the **open challenge**, and is common to both extractors — a characterised domain gap.
- A clear, constraint-compatible path forward: a stronger classifier head and more manipulation diversity, backbone kept frozen.

Thank you — questions welcome.

References

1. Yan et al. *DeepfakeBench*, NeurIPS 2023.
2. Wang & Kalogeiton. *Exposing the Fakes*, ECCV 2024.
3. Ricker et al. *AEROBLADE*, CVPR 2024.
4. Rössler et al. *FaceForensics++*, ICCV 2019.
5. Li et al. *Celeb-DF*, CVPR 2020.